

Study preregistration

Title: The prevalence, reporting practices and reproducibility of pre-study power calculations in sports and exercise sciences research

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1. Background

In Null Hypothesis Significant Testing (NHST), application of the Neyman-Person approach requires us to set in advance the probability of both type I error (alpha level), which is fixed, and type II error by designing a study with adequate power given the true effect size of interest (Dienes, 2008). Power is the probability of finding a true effect and it is determined by the study sample size, the effect size of interest, the study design (i.e., within- or between-subject) and the alpha level (Button et al., 2013; Cohen, 1988; Lakens & Evers, 2014). Conducting studies with adequate statistical power is important because studies with high-power designs have a higher probability of finding a significant effect when and if there is one to be found, and therefore have a higher informational value than studies with low power (Button et al., 2013; Lakens & Evers, 2014). Furthermore, studies with underpowered designs are more likely to produce overestimated effect size estimates after selecting for statistical significance (as is done in most scientific literatures). Thus, an important step when designing an empirical study based on NHST is to justify the sample size by conducting a pre-study power calculation (Lakens, 2022).

Despite the importance of power in the Neyman-Pearson approach to NHST there are suggestions that many sport and exercise science researchers design and conduct studies without performing a pre-study power analysis (Abt et al., 2020; McKay et al., 2022). Abt et al., (2020) reported that only 12 out of 120 (10%) studies published in the *Journal of Sports Sciences* reported a pre-study power analysis. Similarly, McKay et al., (2022) reported that only 84 out of 635 (13%) studies published in motor behavior research conducted a pre-study power calculation. Lack of pre-study power analyses is specially concerning in fields where typical sample sizes are usually small. For instance, Abt et al., (2020) reported a median sample size of 19 participants in the 120 studies they reviewed. Although it is possible that studies with small sample sizes have sufficient power, for example if the authors have good reasons to expect a very large effect, the combination of a lack of a-priori power analyses and small sample sizes is a possible cause for concern.

Researchers might also resort to the practice of calculating the power of the study design after the study has been completed, which is known as a post-study or retrospective power analysis. A post-study analysis uses the study sample size and the effect size observed in the given study, along with the assumed type I error rate to calculate the observed power. As opposed to a pre-study power analysis, a post-study power analysis is considered poor practice (Christogiannis et al., 2022; Maxwell et al., 2008; Yuan & Maxwell, 2005). Despite the irrelevance of power at the analysis stage, researchers sometimes calculate the post-study power, especially when a non-significant effect has been observed (Christogiannis et al., 2022; Maxwell et al., 2008; Yuan & Maxwell, 2005). To what extent this poor practice is observed in the field of sports and exercise science is unknown.

If pre-study power analyses are reported, there is some evidence that they are often irreproducible (Abt et al., 2020; McKay et al., 2022). For instance, Abt et al., (2020) reported that all 12 studies that included a pre-study power analysis failed to include full information on the statistical test(s) to be conducted to detect the chosen effect size, and 4 failed to include a convincing rationale for why the given effect size was chosen. This is key information that should be included in a sample size justification (as well as a specification of the dependent variable, the intended sample size, the statistical test, the alpha level, the effect size type, the effect size magnitude, and the intended power) because the statistical power depends on the chosen statistical test and the magnitude of the effect size of interest. The omission of all required information that a pre-study power analysis is based on indicates that researchers attempting to reproduce these study power analyses might face some difficulties when evaluating the quality of the power analysis. The extent to which these observations about the quality of reporting practices of pre-study power analyses is representative for sports and exercise science as a field remains to be determined.

When researchers plan a *new* study and use a pre-study power analysis, they may use an original/previous study effect size as the basis for sample size planning (Anderson et al., 2017; Lakens, 2022). However, researchers may overlook that studies with underpowered designs and with the presence of publication bias are likely to overestimate true effect sizes, resulting in overly optimistic results (Anderson et al., 2017). A researcher who thus attempts to obtain a power of 80% in his or her study based on these previous studies, may have an *actual* power lower than 80% because of the overestimation of the true effect size in the original study (Anderson et al., 2017). Researchers should take into account that pre-study power analyses based on effect sizes obtained from the scientific literature could be overly optimistic.

2. Research Questions

Given the importance of using valid pre-study power analyses to design informative studies, and the need to provide sufficient details about them so that peers can evaluate the quality of the pre-study power analyses, the aim of this intended research is to quantify the proportion of studies that include a pre-study power analysis in a sample of 350 studies published in 10 sports and exercise science journals. Furthermore, based on previous findings (Abt et al., 2020; McKay et al., 2022), we also aim to report on the quality of the reporting practices and reproducibility of the corresponding pre-study power analyses. By doing so, we attempt to answer the following research questions (RQs):

RQ1: What proportion of studies include a pre-study power analysis?

RQ2: What proportion of studies include a post-study power analysis?

RQ3: Does the dependent variable reported in the pre-study power analysis match the key dependent variable stated in the study hypothesis?

RQ4: Is there a difference in the proportion of studies with a pre-study power analysis that find evidence for the primary hypothesis in comparison to studies without a pre-study power analysis?

RQ5: How often do pre-study power calculations report the dependent variable, the intended sample size, the statistical test, alpha level, effect size type, effect size magnitude, effect size justification and the intended power?

RQ6: If a study with a pre-study power analysis justified the effect size based on a previous study, what was the sample size of the original study?

RQ7: Is there a difference in sample sizes between studies that conducted a pre-study power analysis and studies that did not?

RQ8: what proportion of pre-study power calculations reported in our sample of studies can be reproduced using only the information reported in the article?

RQs 1, 2, 3, 5, 6 and 8 are estimation questions and will not be answered by conducting a hypothesis test. The primary research question is **RQ1** so we will conduct a precision analysis to estimate the sample of studies required to answer such question.

For **RQs 4 and 8**, we will conduct two secondary hypothesis tests for which we will only control for type I error but not for type II error. This is because the size of the effect is unknown and therefore, we cannot perform a pre-study power analysis.

- a) we will explore whether studies with a pre-study power analysis find evidence for the primary hypothesis more often than studies without a pre-study power analysis
- b) we expect that there will be a statistically significant difference in sample sizes between studies that conduct a pre-study power analysis and studies that do not include it.

3. Methods

The pre-registration of our study occurs prior to collecting and coding of data. This preregistration is part of a research project consisting of 3 studies, which will be preregistered all at once, where we aim to survey a large sample of studies from sport and exercise science literature to investigate:

- a) Prevalence, reporting practices and reproducibility of pre-study power analyses (current chapter);
- b) Assessing the presence of publication bias and studies with underpowered designs by conducting a z-curve analysis (chapter 4); and
- c) Reproducibility of p -values and reporting practices (chapter 6)

The data for these 3 preregistrations will be coded over one single data collection. As a result, the same sample of studies will be used in each of the 3 preregistered studies. Deviations from this original protocol will be explicitly acknowledged and documented in any final report of this study.

3.1. Study design

This is a retrospective observational survey study.

3.2. Study sample size

Given our primary research question (**RQ1**), we conduct a precision analysis to estimate the number of studies required to find a *true* proportion of 30% for studies that include a pre-study power calculation in the field of sports and exercise science. To date, two studies in the field of sports and exercise science (Abt et al., 2020 and our pilot study, see <https://doi.org/10.51224/SRXIV.267>) and one in the field of motor behavior (McKay et al., 2022) have reported that 10% (Abt et al., 2020; 12 out of 120 studies), 26% (pilot study) and 13% (McKay et al., 2022; 84 out of 635 studies) of studies included a pre-study power calculation. Based on the previous results, and

the fact that the maximum sample size obtained in a precision analysis is obtained with a proportion of 50% (the larger the proportion up to 50%, the larger sample of studies), we have decided to use an estimate of 30% as the *true* proportion of studies that include a pre-study power analysis in our field.

A precision analysis enables one to examine how precision (specifically confidence interval margin of error) varies with sample size for a given effect size (Gelman & Hill, 2006; Rothman & Greenland, 2018). To perform a precision analysis, one can take the relevant confidence interval formula, and specify two of the following parameters, solving for the third: effect size, sample size, and margin of error. Additionally, one can plot a precision curve (see **Figure 1**; see also **Table 1**) to illustrate the relationship between sample size and precision across a range of values. The R script used to conduct these calculations can be found at osf.io/sample_size_calculation. Using 0.5 as the expected proportion is appropriate in situations where one does not know what to expect because it leads to a maximal sample size estimate (Gelman & Hill, 2006, p. 442). We are interested in estimating proportions, so we use the following confidence interval formula (Gelman & Hill, 2006, p. 442):

$$MOE = z \cdot \sqrt{\frac{p(1-p)}{n}}$$

where MOE = margin of error for the 95% confidence interval, z = z-score corresponding to the desired confidence interval (e.g., 1.96 for a 95% confidence interval), p = expected proportion, and n = sample size.

If we solve for n , the formula is:

$$n = \left(\frac{\sqrt{p(1-p)} \cdot z}{MOE} \right)^2$$

Thus, assuming that $MOE = 0.05$, $z = 1.96$ and $p = 0.30$, a sample size of 323 studies is obtained. Yet, a sample of 350 studies (35 studies per journal) will be collected allowing for a MOE of 0.048 (**Figure 1**; and see also **Table 1**).

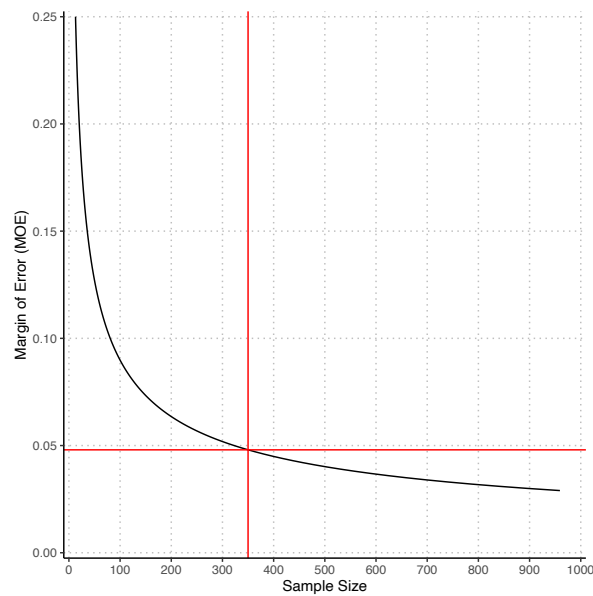


Figure 1 Precision curve showing margin of error for a 95% confidence interval as a function of sample size assuming a proportion of 0.3.

Table 1 Sample size required to obtain a proportion of 30% given a 95% CI and a range of margin of error (MOE).

MOE	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.0	0.05	0.05	0.05
	1	2	3	4	5	6	7	8	9	5	1	2	3
Sample size	480	457	436	417	398	381	365	350	336	323	310	298	287

MOE = margin of error

3.3. Journal and study selection protocol

The study selection protocol is based on the *Selection Protocol for Replication in Sports and Exercise Science* (Murphy et al., 2022). Only applied sport and exercise science studies in the subdisciplines of physiology, sports performance, physical activity, injury prevention and psychology will be considered. By applied sport and exercise science studies we refer to the study of changes in human performance in response to physical activity, exercise and sport. This protocol excludes reviews of any type (e.g., systematic, narrative, scoping and educational), meta-analysis, consensus statements, opinion pieces or commentaries, editorials, case studies, conference proceedings, study protocols, perspectives, and methodological reports. Moreover, only confirmatory studies will be considered. By confirmatory we refer to any study that tests a hypothesis with an experimental or quasi-experimental design. A quasi-experimental design is defined as an empirical study used to estimate the causal impact of an independent variable on target population without randomization. Furthermore, studies which contain single inference tests such as the *t*-test and *F*-tests will be included. For *F*-tests, both main effects and interaction effects will be considered. Although we attempt to apply the protocol described in Murphy et al., (2022) to select studies, we will deviate on two points from the original protocol. First, while the *Selection Protocol for Replication in Sports and Exercise* only selected studies that reported a statistically significant main effect, we will consider both statistically significant and nonsignificant effects. Second, we will also consider interaction effects as opposed to the original selection protocol.

Based on *Selection Protocol for Replication in Sports and Exercise Science*, 10 journals that publish applied research in sports and exercise science and that are ranked in quartile 1 according to www.scimagojr.com (as of 13th September, 2022) will be screened. The inclusion of these journals is based solely on the aims and scopes provided on the journal website where it includes any of the disciplines of applied sports and exercise science that have been described above. Although the *Journal of Physiology* is ranked in quartile 1 (see www.scimagojr.com), such journal publishes very few applied studies. Alternatively, *Journal of Applied Physiology* (quartile 2) will be included instead because it publishes a higher proportion of applied studies. Thus, the 10 journals that will be included are the following: *International Journal of Sports Physiology and Performance*, *Medicine and Science in Sports and Exercise*, *Journal of Applied Physiology*, *Journal of Science and Medicine in Sport*, *Journal of Strength and Conditioning Research*, *International Journal of Sport Nutrition and Exercise Metabolism*, *Scandinavian Journal of Medicine and Science in Sports*, *Psychology of Sport and Exercise*, *European Journal of Sport Science* and *Journal of Sports Sciences*. Given our sample size of 350 studies, 35 studies for each of the 10 selected journals will be screened.

3.4. Search strategy

To identify whether the study included a pre-study power calculation or not, we will search for the following keywords in text: “power”, “sample”, and “size”. We will also screen the methods section in both the “Participants” and “Statistical Analysis” sections.

3.5. Inter-rater reliability

Prior to collecting any data, and because we anticipate that there might be a wide range of hypotheses, study designs and reporting language in the sampled studies, we will develop our coding strategy over a 3-step process. First, the four authors (C.M., J.M., D.L. and J.W.) will meet to discuss and develop the coding strategy created by C.M. Ambiguities on the selection protocol will be discussed and any amendments will be made. Second, three coders (C.M., J.M. and J.W.) will independently code a randomly selected subset of 30 studies from the sample pool of studies (350) as a pilot study. After collecting the data for each of these 30 studies, the three coders will compare the coded data and any disagreement in the parameter selected will be discussed and used to improve the coding strategy. Thirdly, the same three coders (C.M., J.M. and J.W.) will independently code a second randomly subset of 30 studies to assess inter-coder reliability. Inter-coder reliability will be assessed by calculating a pooled Cohen’s kappa estimate based on reliability assessments of each coding category. Any discrepancies in coding decisions will be resolved through discussion and used to improve the coding process. Any modification made to the protocol after assessing inter-rater reliability will be disclosed.

3.6. Data extraction

To minimize errors and ensure quality of data, we will use independent dual coding whereby each sample study is coded twice. Thus, both J.M. and J.W. will each code 175 studies, and C.M. will code the full sample of studies (350). Any discrepancies in coding decisions will be resolved through discussion. D.L. will provide guidance when discrepancies arise and agreement between two coders cannot be reached. C.M. will sample and download 350 articles in pdf, which will then be randomly allocated to the coders. Each coder will independently screen each article and data will be collected on an online survey using Google forms (docs.google.com/survey).

3.7. Coding strategy

Here we provide a description of each of the steps that will be followed to collect data below, which is summarized in **Table 2**.

1. **Does the study report a pre-study power analysis?** This refers to the procedure used to estimate the sample size required to achieve a certain level of statistical power given an effect size of interest, statistical test and alpha level before data collection begins. It is considered a good research practice because allows researchers to ensure that the study has a high probability of hypothesis test to detect a statistically significant effect when H_0 is false. Coders will answer this question by answering *yes* or *no*.
2. **If the study does not include a pre-study power analysis, does it include a post-study power analysis?** A post-study power calculation is conducted once the data has been collected and it is considered a bad research practice. Power calculations are usually reported in the Participants or Statistical Analysis section. Note that post-study power analyses can also be reported directly on the Results without a description in neither Participants nor Statistical Analysis.
3. **Pre-study power analysis statement quote:** this refers to the statement where authors report the parameters used to conduct the pre-study power analysis of their study.
4. **Does the pre-study power analysis statement include a dependent variable?** here we pretend to answer the question whether the dependent variable stated is a clear and specific variable or otherwise is an “umbrella term”. By “specific” we refer to a construct that has been clearly operationalized into a specific and clear variable. By “umbrella term”, we refer to a dependent variable that can be operationalized in several alternative ways. For instance, an “umbrella term” would be when authors state “endurance performance” as a dependent variable in the pre-study power analysis but then this variable is operationalized as maximal oxygen uptake, critical power and maximal aerobic power. To answer this question, coders will compare the dependent variable stated in the pre-study power analysis to listed dependent variables reported in Abstract and Results. If the dependent variable stated in the pre-study power analysis matches any dependent variable listed in the Abstract or Results, coders would answer that the dependent variable is “clear and specific”. Alternatively, if coders notice that the dependent variable is operationalized into several measures, the coder will answer that it is an “umbrella term”. In case that the pre-study power analysis does not include a dependent variable, we will screen the abstract and hypothesis/aims to know whether the study states the key dependent variable. If no specific and clear dependent variable is reported, coders will report that no dependent variable has been reported– that is, “no” as an answer.
5. **Dependent variable:** this refers to the dependent variable stated in the pre-study power analysis and for which the effect size of interest is related to.

6. **Does the dependent variable in the pre-study power statement match any of the variables stated in the hypothesis?** The coder will evaluate whether the dependent variable reported in the statement of the pre-study power analysis matches any of the dependent variables stated in the hypothesis statement. This question will be answered with a *yes*, *no*, or *unclear*. In order to evaluate whether the dependent variable stated in the statement of the pre-study power analysis matches the one reported in the hypothesis statement or not, coders should be aware that dependent variables can be operationalized as follows (Wicherts et al., 2016):

- 1- **Using alternative measures of the same variable:** this refers to when a dependent variable is scored or measured in various alternative ways. For instance, second lactate threshold (LT₂) is operationalized in three alternative ways such as power output at LT₂, heart rate at LT₂ and blood lactate at LT₂. In case that the dependent variable has been measured at different speeds, power outputs or timepoints, every measure would be considered as an alternative measure of the dependent variable. In this case, we would consider that LT₂ has been operationalized in three alternative ways.
- 2- **Using different constructs of the same phenomena:** this refers to when the dependent variable is measured in several alternative ways. For instance, “endurance performance” is measured as peak power output, incremental time to fatigue, lactate threshold test. Another example could be that a study uses as a dependent variable “Functional Movement Screen” (FMS), which is a screening tool used to evaluate *seven fundamental* movement patterns. In such case, the authors from this study did not specify which FMS test was used and only reported FMS as a dependent variable.

In the first case, coders would report that the dependent variable stated in both the hypothesis and the pre-study power analysis match. In the second case, coders would report that they did not match.

7. **Intended total sample size:** this refers to the sample size needed to achieve the intended power, if stated. Coded as NA if not provided.
8. **Statistical test for power calculation:** this refers to the statistical test used to compute the pre-study power calculation. Coders will choose one option among: *paired t-test*, *unpaired t-test*, *one-way within-subject ANOVA*, *one-way between-subject ANOVA*, *two-way within-subject ANOVA*, *two-way between-subject ANOVA*, *two-way mixed ANOVA*.
9. **Alpha level:** this refers to the alpha level used to conduct the pre-study power calculation. Coders will choose one option among: *two-tailed at 0.05*, *one-tailed at 0.05*, *one-tailed at 0.025*, *two-tailed other*, *one-tailed other* and not reported.
10. **Effect size estimate:** this refers to only standardized effect size estimates (see code 11 below) used to conduct the pre-study power calculation. Coders will report the effect size estimate or otherwise not reported.
11. **Effect size type:** coders will choose one option among: d_z , d_s , d_{pop} (Cohen’s d family); f (Cohen’s f); g , g_{av} , g_{rm} (Hedge’s g family); Glass’s Δ ; η^2 (eta squared); η_p^2 (eta partial squared); ω^2 (omega squared); ω_p^2 (partial omega squared); unstandardized effect size (i.e., mean difference) and not reported.

12. **Justification of the effect size of interest:** this refers to the justification of the effect size that researchers are interested in. Coders will choose one option among: *previous study*, *pilot study*, *meta-analysis*, *Smallest Effect Size of Interest (SESOI)*, *Minimal Statistically Detectable Effect Size (MSDES)*, *Cohen's d thresholds* and *no justification*. The definition of each one of these options is as follows:
- **Previous study:** researchers select an effect size reported in a previous published study.
 - **Pilot study:** researchers perform a pilot study and use the effect size from the pilot study to determine what the sample size in a follow-up study should be based on a pre-study power calculation.
 - **Meta-analysis:** researchers select a meta-analytic effect size reported in a meta-analysis.
 - **Smallest Effect Size of Interest (SESOI):** researchers select the smallest effect that would be practically relevant, theoretically predicted, or that would reject a previously observed effect using the Small Telescopes approach. The goal is to determine the smallest increase in a relevant outcome measure that is subjectively deemed to be large enough to matter (Anvari & Lakens, 2021).
 - **Minimal Statistically Detectable Effect Size (MSDES):** researchers calculate the smallest statistically significant effect size estimate given a sample size. For instance, researchers may only be able to recruit 50 participants, so they could conduct a pre-study power calculation to estimate the smallest statistically significant effect size given a sample size of 50 and an intended power of 80%.
 - **Cohen's d thresholds:** researchers select an effect size based on Cohen's d thresholds ($d \leq 0.2$, small; $0.2 < d < 0.8$, medium; $d \geq 0.8$, large).
 - **No justification:** researchers do not provide any justification whatsoever about the effect size of interest.
13. **If the effect size justification was based on a previous study, what is the sample size of the original study?** Coders will report the sample size of the study from which the effect size was obtained.
14. **Intended power:** this refers to the power that the study aims to achieve given an effect of interest, sample size and alpha level.
15. **Statistical software:** this refers to the statistical software used to conduct the pre-study power calculation. Coders will choose one among: *G*Power*, *R*, *SAS*, *other*, and *not reported*.
16. **Others:** this refers to any other information reported in the statement of the pre-study power calculation that is required for reproducibility purposes and that it is seldom reported. For instance, parameters such as sphericity correction, correlation between measurements, number of groups and number of measurements are required for factorial ANOVA tests.
17. **Does the statistical test reported in the pre-study power statement match any of the statistical tests reported in the Statistical Analysis section?** Coders will evaluate with a yes, no or unclear whether the statistical test reported in the statement of the pre-study power calculation matches any of the statistical tests reported in the Statistical Analysis section. In case that one of the reported statistical tests is a t-test and the other one is one-way ANOVA with two factors, we will consider that both statistical tests match as they are statistically equivalent.

18. **Achieved sample size:** this refers to the study sample size reported in the current study (not in the statement of the pre-study power calculation) and used to conduct the statistical analysis.
19. **Does the study support the research hypothesis?** determining whether the authors of a study concluded the results supported their hypothesis involved three steps:
1. The primary hypothesis of a study is identified, either because the authors specify the hypothesis as primary, it is the study hypothesis that includes the dependent variable reported in the pre-study power analysis or it is the hypothesis most applicable to applied sport and exercise science research.
 2. If the pre-study power analysis includes a specific and clear dependent variable, we will use this one to determine whether the study supports the hypothesis. If the dependent variable is an “umbrella term”, coders will select a clear and specific dependent variable listed in that order in: 1) hypothesis; 2) aim; 3) abstract; 4) results.
 3. In the absence of a pre-study power analysis, and in the case that multiple dependent variables are stated in the hypothesis, coders will select a clear and specific dependent variable listed in that order in: 1) hypothesis; 2) aim; 3) abstract; 4) results.
 4. We will code support for hypotheses based on the interpretations in each paper, not based on our own criteria. Thus, if the authors predicted no effect of an independent variable, observed null results, and then concluded the results supported their hypothesis, we coded this as support for the hypothesis.

3.8 Reproducibility of pre-study power calculations

The authors C.M. and J.W. will attempt to reproduce all pre-study power analyses. The authors will begin by attempting to calculate the power calculation using the parameters that were reported in the study. A pre-study power analysis will be deemed as *reproducible* if the sample size could be confirmed using the reported parameters, otherwise it will be deemed *non-reproducible*. If insufficient parameters are explicitly reported, which we assume will be likely based on previous findings (Abt et al., 2020; McKay et al., 2022), we will attempt to make assumptions for missing parameters. For example, we expect that most studies using factorial designs will not report the number of groups and measurements as input parameters required to conduct a pre-study power calculation. We also expect that other parameters such as the correlation between measurements or the non-sphericity correction parameters will not be reported often. If the statistical analysis is not reported, we will assume they used the same model as the actual analyses reported in the Results of the study. All plausible analyses will be attempted, but effect size, intended power, and alpha will not be guessed. Studies that can be reproduced by assuming parameters will be recorded as *conditional*, otherwise they will be considered *non-reproducible*. If results disagree with those reported in the original pre-study power analysis, we will report the difference between the original and our estimated sample size. The pre-study power analysis will be reproduced with the same statistical software reported in the original study. If the statistical software is not reported, the authors will attempt to reproduce the sample size with G*Power and R.

Table 2 Summary of the coding strategy

Item	Action
1. Does the study report a pre-study power analysis? *If so, go to item 3. Otherwise, go to item 2.	Select from a list (<i>yes/no</i>)
2. Does the study report a post-study power analysis? *Go to item 19.	Select from a list (<i>yes/no</i>)
3. Pre-study power analysis quote	Copy pasted quote of the pre-power analysis
4. Does the pre-study power analysis statement include a dependent variable?	Select from a list
5. Dependent variable reported in pre-study power analysis	Report the dependent variable
6. Does the dependent variable in the pre-study power statement match any of the variables stated in the hypothesis statement?	Select from a list (<i>yes/no/unclear</i>)
7. Intended sample size	Report value
8. Statistical test	Select from a list
9. Alpha level	Select from a list
10. Effect size type	Select from a list
11. Effect size estimate	Report value
12. Effect size justification	Select from a list
13. If the effect size of interest was based on a previous study, what is the sample size of the original study?	Report value
14. Intended power	Report value
15. Statistical software	Select from a list
16. Others	Report additional information
17. Does the statistical test reported in the pre-study power statement match any of the statistical tests reported in the Statistical Analysis section?	Select from a list (<i>yes/no</i>)
18. Achieved (total) sample size	Report value
19. Does the study support the research hypothesis?	Select from a list (<i>yes/no/unclear</i>)

3.9 Statistical analysis

Once the data collection is over, data collected through Google forms will be downloaded in the form of a unique dataset as an excel spreadsheet. This dataset will later be split in three datasets corresponding to each of the preregistered studies. Data will be analyzed in R (R Core Team, 2019).

RQ1: What proportion of studies include a pre-study power calculation? **RQ1** will be addressed descriptively by calculating the percentage of all studies in our sample that conducted a power analysis:

$$\frac{\text{studies with pre-study power analysis}}{\text{studies with pre-study power analysis} + \text{studies without power analysis}} \times 100$$

RQ2: What proportion of studies include a post-study power analysis? **RQ1** will be addressed descriptively by calculating the percentage of all studies in our sample that conducted a power analysis:

$$\frac{\text{studies with post-study power analysis}}{\text{studies with post-study power analysis} + \text{studies without power analysis}} \times 100$$

RQ3: Does the dependent variable reported in the pre-study power calculation match the key dependent variable stated in the study hypothesis? **RQ3** will be addressed descriptively by calculating frequencies.

RQ4: Is there a difference in the proportion of studies with a pre-study power calculation that find evidence for the primary hypothesis in comparison to studies without a pre-study power calculation? **RQ4** will be addressed by performing a two-sided proportions z-test with an alpha level of 0.05 to investigate whether the rate of significant results in studies with a pre-study power analysis is significantly different than in studies without a pre-study power analysis.

RQ5: How often do pre-study power calculations report the dependent variable, the intended sample size, the statistical test, alpha level, effect size type, effect size magnitude, effect size justification, the intended power and statistical software? **RQ5** will be addressed descriptively by calculating frequencies.

RQ6: If a study with a pre-study power calculation justified the effect size based on a previous study, what was the sample size of the original study? **RQ5** will be addressed by calculating descriptive statistics.

RQ7: Is there a difference in sample sizes between studies that conducted a pre-study power analysis and studies that did not? **RQ7** will be addressed by conducting a two-tailed independent samples Welch's *t*-test with an alpha level of 0.05 to test whether studies with power analyses had different sample sizes compared to studies without power analyses.

RQ8: what proportion of pre-study power calculations reported in our sample of studies can be reproduced using only the information reported in the article? **RQ8** will be addressed descriptively by calculating the percentage of pre-study power analysis that could be reproduced (*reproducible* category). We will also calculate the percentage of pre-study power calculations that could be reproduced conditionally by making some assumptions (*conditional* category) as follows:

$$\text{reproducible} = \frac{\text{number of studies with pre-study power calculations reproduced}}{\text{total sample of studies with a pre-study power calculation}} \times 100$$

$$\text{conditional} = \frac{\text{number of studies with pre-study power calculations conditionally reproducible}}{\text{total sample of studies with a pre-study power calculation}} \times 100$$

4. References

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